

Mohammad Rafay

U.S. Citizen | (832) 434-8099 | Richmond, Texas 77407 | rafaymohammad03@gmail.com | linkedin.com/in/mrafay18 | mrafay18.github.io

EDUCATION

Texas A&M University

Bachelor of Science in Computer Engineering

College Station, TX

December 2025

CERTIFICATIONS

CompTIA Information Technology+ (COMP001021854961)

May 2021

EXPERIENCE

Software Engineer

January 2026 – Present

Norstella

Richmond, Texas

- Designed and implemented Java Spring Boot backend services and REST APIs for a healthcare platform, supporting secure client-server workflows, service-to-service communication, and reliable data processing across distributed microservices in a fault-tolerant, scalable backend architecture.
- Optimized SQL queries and backend data access patterns, reducing slow query execution time and improving reliability across REST API workflows.
- Contributed to Agile development workflows by managing code in Azure DevOps repositories using PR workflows with branch policies and build validation, authoring unit tests and performing code reviews that strengthened release stability and code quality.
- Developed, deployed, and tested Java Spring Boot services in an AWS-hosted VM environment, using remote server workflows, API validation, logging, debugging, and operational monitoring to support cloud-based backend development on AWS EC2.
- Applied generative AI development tools (Cursor, Claude, and Forge AI) to accelerate backend engineering workflows, leveraging LLM-assisted code generation, debugging, and validation across Java Spring Boot services.

Software QA Engineer – Career Experience (Messages App)

January 2025 – June 2025

Apple Inc.

Cupertino, California

- Built and optimized reusable test frameworks and automation pipelines for the iOS/macOS Messages App, integrating results into CI/CD reporting cycles to catch defects earlier in the release cycle and support scalable regression testing across software releases.
- Designed and implemented automated tests in Python and Swift using XCTest, developing test cases and coverage strategies at scale to validate UI workflows and verify changes end to end across releases.
- Executed functional, regression, and performance testing; triaged and debugged critical issues to root cause, delivering actionable reports that improved system performance and release stability.
- Partnered with cross-functional engineering teams to maintain test development roadmaps, translating software requirements into testable criteria and driving quality ownership across customer-facing applications.

Software Engineer - Internship

May 2024 – January 2025

Y-12 National Security Complex

Oak Ridge, Tennessee

- Built an interactive route planning simulation using Cesium, Unity, and C#, allowing users to dynamically create the most efficient routes for transporting nuclear materials while ensuring maximum security.
- Implemented cybersecurity policies to strengthen the software for the company's alarm response training, enhancing the facility's protection against imminent threats by 40%.
- Accelerated training throughout the organization by 20% developing modules to counteract cyber threats in resource transportation.

Cybersecurity Engineer - Internship

May 2023 – August 2023

Texas Department of Transportation

Austin, Texas

- Maintained Security Policy content for compliance and accuracy throughout the organization.
- Conducted Third Party Risk Management assessments for enhanced protection of management systems.
- Utilized KQL within Microsoft Azure, to analyze research metrics, supporting data-driven decision-making and strategy formulation.

PROJECTS

Accent Auto (Client Website) | HTML, CSS, JavaScript, Node.js, Express | accentautohtx.com

January 2026 – Present

- Built a client website using HTML, CSS, JavaScript, Node.js, and Express, implementing responsive inventory pages, vehicle detail views, status filters, and customer contact and inquiry workflows including form submission and email integration.

BluStick | TypeScript, JavaScript, Swift, C++, SQL

August 2025 – December 2025

- Developed a hardware-software system that collects Bluetooth signals from distributed sensors, performs real-time triangulation, and visualizes device locations on an interactive and responsive cross-platform user interface.

LED Matrix Camera Display | Python

January 2024 – May 2024

- Built an interactive LED matrix system that streams live webcam input and button-triggered patterns, validating UI responsiveness and ensuring seamless interaction between input devices and visual output.

Point-of-Sale System | TypeScript, JavaScript, Java, HTML, React

January 2024 – May 2024

- Developed a React and TypeScript point-of-sale application with role-based employee, manager, and cashier interfaces, enabling order management, transactional payment flows, and structured front-end user-interface.

SKILLS

Languages: Python, Java, C++, C#, JavaScript, SQL, TypeScript, Swift, Bash, HTML, CSS

Technologies: Spring Boot, REST APIs, React, Node.js, Express, Git, Azure DevOps, CI/CD, XCTest, KQL, AWS EC2, Linux/Unix, Postman, Agile, JIRA, Cursor AI, Claude

Soft Skills: Communication, Cross-Functional Teamwork, Problem Solving, Analytical Thinking, Time Management, Leadership

LEADERSHIP POSITIONS

President | Corporate Chair | Honors Society for Electrical & Computer Engineers (HKN)

January 2023 – January 2025